# Testing Low-Rank Causal Control in AdamW Memory

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## Abstract

Optimizer state can alter future learning without changing a network's current predictions. We study whether low-rank projections of AdamW first-moment memory preserve the functional effects of history-derived interventions. Our evaluation separates independent-image operator capture, signed-response reconstruction and robustness across future batches and horizons. A frozen thirty-task confirmation fails its declared causal-sufficiency conjunction. Subsequent diagnostics identify partial short-horizon gains from fitting the response operator to the desired future. Under deteriorating continuation, these gains transfer poorly. An eight-history, accuracy-stable CIFAR10 follow-up instead improves both futures descriptively: rank 16 sixteen-update error falls from 0.965 to 0.838 on the fitted future and from 0.924 to 0.850 on an alternate future, while the declared interaction test fails ($p = .145$). At sixty-four updates, rank 16 errors exceed zero-response prediction in every history on both futures; rank 64 does so in eight fitted and seven alternate futures. Response geometry characterizes the discrepancy: substantial retained magnitude coexists with weak directional alignment. These results support horizon- and protocol-specific evaluation of optimizer-state representations, rather than certifying causal sufficiency from a fitted spectrum or short-horizon gain.

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## 1 INTRODUCTION

Adaptive optimizers retain information beyond model weights. Consequently, two networks with identical current predictions can learn differently because their moment states differ. This makes optimizer state a candidate representation of training history. The practical scientific question is whether a compact representation preserves what that history *does* to later learning, rather than merely approximating its coordinates or its immediate influence.

An earlier thirty-task susceptibility diagnostic makes the evaluation hazard concrete. A history-derived direction was 90.93× as potent as the median of eight random directions, yet a separate scalar predictor that improved mean squared error over a rich observational comparator by about 15.7% was about 2.2% worse than predicting zero response. These are susceptibility and scalar-prediction results, not low-rank preservation results; we use them only to motivate the stricter tests below (Appendix B).

A low-rank response spectrum offers a plausible starting point, but leaves three questions open. Does a fitted basis capture influence on independent images? Does projecting a history-derived moment displacement preserve its actual signed effect? Does that preservation persist under a different future batch sequence and a longer horizon? These requirements concern different distributions and estimands. Treating one as evidence for the others can make an apparently compact representation look causally sufficient before it has been tested.

We test first-moment interventions in AdamW (Loshchilov and Hutter, 2019), holding weights, second moments and clock fixed. Three comparisons supply the central evidence. A frozen discovery–confirmation study rejects a declared conjunction of capture, historical alignment and finite causal sufficiency. Equal-rank immediate versus horizon-matched operators then show that including the desired future can improve reconstruction. Finally, an eight-history

stable CIFAR10 follow-up separates a short-horizon gain from longer-horizon preservation. Matching helps both futures descriptively, while its declared interaction test fails; at H64, retained response magnitude is accompanied by poor directional alignment and reconstruction often worse than predicting zero.

The contribution is a diagnostic protocol and an empirical boundary on what the tested response subspaces preserve. Exact finite-anchor and ResNet-GN/ViT diagnostics support the horizon distinction; they are not extra independent replications of the same experiment. Earlier alternate-future failure is conditional on a deteriorating continuation regime and is not reproduced by the stable follow-up. This correction matters: the strongest supported conclusion concerns the distinction between influence capture and durable effect preservation, rather than a universal failure of transfer.

The study is deliberately bounded. Capture remains modest even at rank 64; histories per natural-data cell number four or eight; and the stable CIFAR100 cohort is incomplete. We evaluate perturbation-preservation in a fixed Euclidean moment chart, not a deployable optimizer compressor or a controller with established accuracy benefits. These boundaries specify both the use of the diagnostic and the hypotheses it leaves open.

### 1.1 Relation to Previous Work

Differentiation through momentum-based training predates this study (Maclaurin et al., 2015). Recent work studies hidden learning-state responses (Wang, 2026), first-moment transport and future-dependent valuation (Xu, 2026), and full-trajectory versus immediate sensitivities in short-horizon training decisions (Guo, 2026). Our derivative construction applies these established ideas. The empirical question here is whether an operator-derived subspace preserves a held-out history-derived effect, and whether that preservation survives changing the readout horizon and committed future.

Low-dimensional adaptive methods also address statistic transport, as in LDAdam (Robert et al., 2025); independent-batch subspace identifiability has been examined in low-rank training (Thomas, 2026). These methods motivate compact optimizer representations, but their training objectives differ from our intervention-preservation estimand. We therefore make no algorithm-performance comparison to them. Our comparisons separate independent output-anchor capture from actual signed-response reconstruction, compare immediate and horizon-matched bases at equal rank, and evaluate the same fitted basis under an alternate future.

## 2 INTERVENTIONS AND ESTIMANDS

Let $s = (\theta, m, v, t)$ be a saved AdamW state with $p$ first-moment coordinates. For a committed sequence of future batches and augmentation randomness $\omega$, write $T_\omega^H(s)$ for $H$ updates. All intervention branches within a future share this sequence and the unchanged recipient state.

**Functional chart.** For a bank $B$ of $N$ images and $C$ classes, let $Z_B(\theta) \in \mathbb{R}^{N\times C}$ be the logit matrix. Let $Q \in \mathbb{R}^{C\times(C-1)}$ have orthonormal columns orthogonal to the class-common direction. For $(\theta_H, \ldots) = T_\omega^H(s)$, define

$$f_{H,B,\omega}(s) = \frac{\mathrm{vec}(Z_B(\theta_H)Q)}{\sqrt{NC}}. \tag{1}$$

This chart removes the softmax-invariant common shift. Its norm equals the class-centered logit RMS. The full centered chart used in the controlled study gives the same norms and inner products.

**Response operator.** Inject a moment displacement by $Iu = (0, u, 0, 0)$ and define

$$A_{H,B,\omega}u = \frac{d}{d\epsilon} f_{H,B,\omega}(s + \epsilon Iu)\Big|_{\epsilon=0}. \tag{2}$$

An immediate basis uses $H = 1$; a matched basis uses $H = 16$. Leading right directions are fit on calibration images without using the historical direction or response-test images. With orthonormal columns $U_k$, write $P_k = U_k U_k^\top$. Independent capture is

$$\gamma_{H,B',\omega}(P_k) = \frac{\|A_{H,B',\omega}U_k\|_F^2}{\|A_{H,B',\omega}\|_F^2}. \tag{3}$$

The numerator is evaluated by basis-column actions; the denominator is either directly evaluated or estimated with independent Rademacher trace probes, as specified by study. This is operator energy, not historical-direction reconstruction accuracy.

**Finite effects.** For $u = (m_{\mathrm{cut}} - m_{\mathrm{donor}})/\|m_{\mathrm{cut}} - m_{\mathrm{donor}}\|$, use amplitude $\alpha = 0.002\|m_{\mathrm{cut}}\|$. Let $f$ denote the same readout on the independent response bank. We record

$$R_H^{\pm}(u) = f(s \pm \alpha Iu) - f(s), \tag{4}$$

$$D_H(u) = \tfrac{1}{2}\{f(s + \alpha Iu) - f(s - \alpha Iu)\}, \tag{5}$$

$$E_H(P_k) = \frac{\|D_H(u) - D_H(P_k u)\|}{\|D_H(u)\|}. \tag{6}$$

The projected and remainder directions retain their physical norms; they are not renormalized. The historical displacement is a reproducible probe derived

from two saved states, not proof that every injected state is naturally reachable or that this difference isolates a unique causal source. Error one denotes predicting no intervention effect, i.e., using the unchanged continuation as the response prediction; it is not a classification-error baseline. This comparator is required even when a method beats a learned observational predictor. Cosine and norm retention are reported separately because preserving magnitude need not preserve direction. Each sign is also evaluated against the unchanged continuation. Even response and full/projection/remainder additivity diagnose cancellation and local nonlinearity. Responses below the declared RMS floor are unresolved rather than evidence of success.

**Magnitude and alignment.** For nonzero full and projected effects, define $\rho = \|D_H(P_k u)\|/\|D_H(u)\|$ and $c = \cos(D_H(u), D_H(P_k u))$. The exact finite-response identity

$$E_H(P_k)^2 = 1 + \rho^2 - 2\rho c \tag{7}$$

shows why retaining magnitude alone can be misleading. A nonzero projection is worse than zero prediction whenever $\rho > 2c$. We use this elementary identity to interpret saved responses; it is not a new theoretical result.

**Why capture does not certify preservation.** In the linear regime, Eq. 6 becomes $\|A_H(I - P_k)u\|/\|A_H u\|$. Equation 3 averages energy over all directions, whereas this ratio concerns one direction and one future. Neither independent capture nor a fitted spectrum alone supplies that direction-specific guarantee. Appendix A gives the elementary distinction and the full AdamW tangent recurrence. This evaluates direction-agnostic basis selection, not an impossibility theorem about low-rank representations.

# 3 EXPERIMENTAL DESIGN

Table 1 identifies the independent units and each study's role. Successive releases, branches, checkpoints, paths and output images are not independent training replications. The horizon-matched diagnostics follow the original failed confirmation; they are not a redefinition of its outcomes.

## 3.1 Frozen Controlled Test

The controlled tasks render 32 latent factors into $32 \times 32$ images, with four factors determining four-class labels. Each independent task uses 4,096 training examples and disjoint calibration, response and evaluation roles. ResNet18-GN is trained for 500 updates, batch 128; recipient and donor are updates 350 and 200. The operator uses 128 fitting images, 24 range probes, one reorthogonalized subspace iteration and 32 trace probes on each fitting/independent bank. The future-response bank contains 192 images.

Discovery considers ranks $1, 2, 4, 8, 12, 16$. The frozen rule chooses the smallest rank with mean independent capture at least 0.80 and bootstrap lower bound at least 0.70. *No rank qualifies.* The rule therefore freezes rank 16 as a fallback and records failed prequalification. The stronger low-rank headline also requires rank at most eight. Confirmation runs thirty new task identities without changing thresholds or selecting seeds. Appendix B reports the principal gates and outcomes. These are internally frozen protocols, not an external preregistration. Capture and sufficiency thresholds define the original operational claim; they are not universal cutoffs for useful memory. Rank 16 is its failed-prequalification fallback, and rank 64 is a secondary budget expansion, rather than a post hoc successful rank selection.

## 3.2 Natural-Data Diagnostics

CIFAR10 and CIFAR100 (Krizhevsky, 2009) supply official training and test splits. The exact-anchor study trains a small ConvGN for 2,000 updates and computes finite fitting-bank rows directly. The architecture extension uses width-32 ResNet18-GN (He et al., 2016; Wu and He, 2018) and a six-block ViT with width 192, three heads and $4 \times 4$ patches. It trains 8,000/12,000 updates on CIFAR10/100, with early/late checkpoints and two futures per checkpoint. Its primary basis uses 512 fitting images, 128 probes and two subspace iterations. Alternative fitting packages diagnose estimator stability.

The small-model matching study reuses four saved exact-anchor states. The fresh matching study trains eight new ResNet-GN histories, four per dataset, for 8,000 updates with donor at 5,600. Its 2.80/2.82-million-parameter models use GroupNorm and SiLU (Elfwing et al., 2018). It reserves 4,096 training images for disjoint fitting, independent-capture and calibration roles. The nested fitting banks contain 64/256 images; the held-out capture bank contains 256. Response tests use 128 official test images; endpoints use the remaining 9,872.

The immediate and matched bases use 128 randomized range columns and one stabilized power iteration (Halko et al., 2011); rank 16 is primary and rank 64 diagnostic. On each dataset's first seed, both matched banks are compared with 192 columns. Fit-energy relative change must be at most 10%, and projected-response difference/full-response norm at most 15%.

Table 1: Evidence structure. Numbers of histories are independent training units; repeated cases are nested within them. Exactness refers to the finite fitting bank. All tested moment displacements leave the current model function unchanged.

| Study | System | Independent units | Question addressed |
|---|---|---|---|
| Frozen confirmation (07) | Controlled four-class tasks; ResNet18-GN | 20 discovery, 30 confirmation | Does the declared conjunction pass? |
| Exact-anchor diagnostic (10) | MNIST/MLP and CIFAR10/ConvGN | 4 histories per dataset | Is approximation the sole explanation? |
| Architecture extension (11) | CIFAR10/100; ResNet18-GN and ViT | 4 histories per cell; 16 total | Does immediate-to-future loss recur? |
| Small-model matching (12) | CIFAR10/ConvGN; reused states | 4 histories reused from 10 | Does matching the horizon help? |
| Fresh-history matching (13) | CIFAR10/100; ResNet18-GN width 32 | 4 fresh histories per dataset | Does fitted-future gain transfer? |
| Stable follow-up (14) | CIFAR10; ResNet18-GN width 32 | 8 fresh histories | Does matching remain context specific under stable continuation? |

These checks qualify a limited estimator comparison, not exact global singular vectors.

Only path 0 fits the matched basis. Path 1 shares the first 128-image unaugmented calibration update and then diverges. Later future batches contain 32 augmented examples drawn with replacement; this differs from training batch 128. Thus "alternate" means transfer after a shared first step. The matched basis is an oracle for its declared future, not an online controller that must predict future data. Full hyperparameters and compute records are in Appendix C.

### 3.3 Stable-Continuation Follow-Up

Study 14 freezes a new protocol after the earlier results. Eight fresh CIFAR10 histories use the same architecture, 8,000 training updates and donor at 5,600. Continuation keeps batch 128 and lowers learning rate from .001 to .00012. Before intervention analysis, every history and both futures must retain accuracy within two percentage points and NLL within .10 of the cut at H1/H16/H64; no histories are excluded or retuned. All eight pass. Fitting/capture banks contain 256 images; response and endpoint banks contain 128 and 9,872 official test images. Rank 16 is primary; rank 64 is diagnostic. Wider 192-column checks on the first two histories use the frozen 10% energy and 15% projected-response tolerances. Both futures share the first calibration step. These changes constitute a combined protocol follow-up, not an isolated batch-size or learning-rate ablation.

### 3.4 Inference and Validation

The fresh study's primary endpoint is CIFAR10, path 0, horizon 16, rank 16 and fitting bank 256; CIFAR100 is a separate replication cohort. Alternate futures, bank growth, rank 64 and other horizons are secondary diagnostics. Figures show seed-level measurements or means with minimum–maximum ranges. Four positive differences among four independent histories have one-sided sign tail $1/16 = 0.0625$; branch counts cannot increase this sample size. We report descriptive evidence rather than broad significance.

For study 14, the declared primary statistic is the within-history fitted-minus-alternate matching gain at H16/rank 16. Its one-sided sign test retains $\alpha = .025$, the original per-dataset threshold for two planned cohorts, despite only CIFAR10 being completed. For each future, matching gain is immediate-basis error minus matched-basis error; the primary interaction subtracts the alternate-future gain from the fitted-future gain. Separate path gains and H64 are secondary. This manuscript reports the completed stable CIFAR10 diagnostic. The planned stable CIFAR100 replication was unfinished at the reporting cutoff and contributes no observations. No stopping or seed selection is based on effect estimates.

Qualification includes native/functional gradient agreement, tangent/adjoint duality, and finite native differences at full/half amplitude. Both signs share batches and RNG with the baseline. Saved-output arithmetic is checked separately from neural execution. Archive hashes verify supplied bytes; they are not independent execution authentication.

Table 2: Larger-model extension, late checkpoints, rank 16. Capture and cosines are means over four histories per cell, averaging repeated paths.

| Dataset / model | Capture | $H1$ | $H16$ | $H128$ |
|---|---|---|---|---|
| C10 / ResNet-GN | 17.50% | .814 | .257 | .015 |
| C10 / ViT | 8.24% | .818 | .315 | .034 |
| C100 / ResNet-GN | 4.79% | .653 | .232 | .019 |
| C100 / ViT | 1.91% | .604 | .267 | .109 |

# 4 RESULTS

## 4.1 The Frozen Conjunction Fails

Confirmation independent capture averages 38.41% (95% task-bootstrap interval 36.69–40.23%). Historical squared projection fraction averages 0.0443, below the declared 0.15 alignment gate. At sixteen updates, projected/full cosine is 0.554 and remainder/full norm ratio 0.842. Both signs fail the baseline-relative sufficiency comparison. Learning competence, locality and nontrivial-effect qualification pass; low-rank capture, historical alignment and causal sufficiency fail. The original strong mechanism claim is therefore unsupported.

A large historical/random potency ratio does not rescue it. Confirmation historical potency is weaker than current-gradient and preconditioned-gradient controls on average. Structured direction potency, subspace membership and finite-response preservation measure different properties.

## 4.2 Immediate Fidelity Does Not Persist

For the four exact-anchor CIFAR10 histories, rank-16 independent capture is 44.12%; projected/full cosine decreases from 0.937 at one update to 0.305 at sixteen and 0.185 at sixty-four. Rank 64 raises immediate cosine to 0.972, but sixty-four-update cosine is 0.196. Randomized fitting error therefore cannot be the sole explanation in this finite-anchor setting.

Table 2 reports the larger extension. Immediate alignment exceeds sixteen-update alignment in all thirty-two checkpoint cases after averaging paths, but these cases belong to sixteen histories. Rank 64 improves intermediate fidelity; its late 128-update mean error exceeds the zero predictor in all four dataset/model cells. No panel reaches the inherited 80% capture target. Budget and anchor checks also leave the fitted bases' stability unresolved, so the study does not identify an authenticated unique control subspace.

## 4.3 Matching Helps the Fitted Future

At sixteen updates, the small-model matched basis reduces rank-16 reconstruction error from 1.352 to 0.816 on its fitted future and from 1.348 to 0.909 on its alternate future. All four reused histories improve on each path. This motivates the fresh-history diagnostic without constituting an independent replication of the original cohort.

In fresh ResNet-GN histories, fitted-path error decreases from 1.049 to 0.594 on CIFAR10 and from 0.965 to 0.797 on CIFAR100 (Fig. 1). All four histories in each cohort improve. The improvements are partial: substantial relative error remains. Matching sixteen updates also sacrifices immediate fidelity; the matched basis is close to zero-effect prediction at one update, while immediate-basis errors are 0.638 and 0.692. Appendix Figure 4 shows this horizon tradeoff.

## 4.4 Limits of Transfer and Operator Capture

On the fresh alternate futures, matched error is 0.993 versus 0.937 for the immediate basis on CIFAR10, and 0.995 versus 0.988 on CIFAR100. Only one history per cohort benefits from matching. This failure concerns the tested basis and context; four seeds cannot establish population equivalence or exclude a small transfer benefit.

Held-out H16 capture at rank 16/fit 256 rises from 4.10% to 39.80% on the fitted CIFAR10 future and from 0.315% to 14.89% on CIFAR100. On alternate futures, matched capture is only 1.27% and 0.116%. Capture corroborates context dependence but does not reach majority energy in either cohort. Larger ranks might preserve more influence; our low-capture result cannot refute a conditional high-capture sufficiency claim.

Rank 64 supplies a useful diagnostic rather than a replacement primary endpoint. Fitted-path errors are 0.447 on CIFAR10 and 0.678 on CIFAR100; alternate errors are 0.958 and 0.991. Improving fitted reconstruction at this larger rank does not restore comparable transfer. Rank 64's numerical status is weaker: the wider-sketch projected-response discrepancy reaches 11.86%.

Increasing the matched fitting bank from 64 to 256 reduces fitted-path H16 error by 0.038 on CIFAR10 and 0.008 on CIFAR100, versus approximately 0.001 and 0.0001 on alternate futures. This specific image-support expansion does not restore transfer. It does not rule out other ranks, banks or representations.

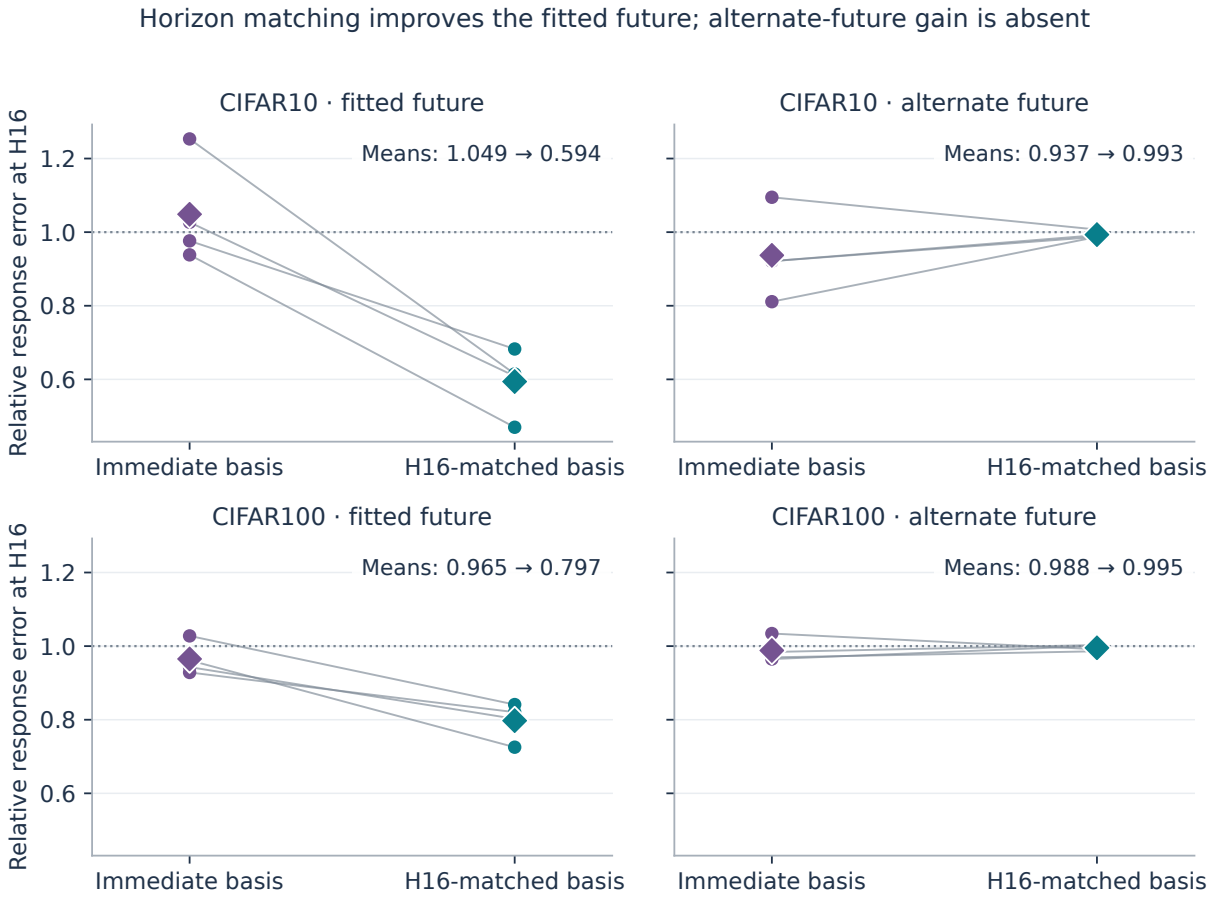



Figure 1: Study 13: equal-rank reconstruction at sixteen updates, fitting bank 256. Points are four independent histories per dataset; lines connect paired measurements and diamonds are means. The dotted line is zero-response error. A matched basis improves its fitted future but loses that advantage on the alternate future.

### 4.5 Numerical Validity and Training Quality

Separate positive/negative baseline-relative errors agree with the central-response conclusions. In the fresh study at H16, maximum full-effect even/odd norm ratio is 0.347%, and primary rank 16/fit 256 additivity error is at most 0.306%. The maximum resolved saved-prediction derivative error is 4.44%, below the frozen 15% limit. Unresolved controls remain unresolved. These measurements argue against a large cancelled even response or obvious local derivative breakdown at the tested amplitude; they do not establish universal instrument correctness.

Baseline quality is a separate concern. Cut accuracy is 86.60–87.42% on CIFAR10 and 59.14–60.64% on CIFAR100. After 64 unchanged-optimizer updates, accuracy loses 3.91–11.34 and 13.59–16.13 percentage points, respectively. The future batch-size reduction and constant learning rate are plausible contributors, but neither is experimentally isolated. The paired responses are conditional on these declared futures. They do not demonstrate useful compression during accuracy-stable continuation.

Endpoint intervention effects are much smaller than baseline drift. Across fresh signed branches and controls, maximum absolute baseline-relative accuracy change is 0.111 percentage points and NLL change is 0.002922. There is no prespecified controller with consistent accuracy benefit. At H16, the preconditioned-gradient control has 1.78–7.81 times the historical response norm; one tensor-norm-matched random control has 0.057–0.130 times that norm. Historical potency therefore does not imply a uniquely privileged memory direction.

### 4.6 Transfer Under Stable Continuation

In study 14, all eight histories pass the baseline and saved numerical qualification checks. At H64, unchanged continuation improves endpoint accuracy by 0.91–1.86 percentage points across histories and futures, unlike study 13. Recomputing central effects

from archived branch logits reproduces every saved projection error exactly.

At H16, rank 16 matching reduces mean error from 0.965 to 0.838 on the fitted future and from 0.924 to 0.850 on the alternate future (Appendix Fig. 3). Seven of eight histories improve on each path. The mean interaction is 0.053, positive in six histories; its one-sided sign-test tail is 0.1445, so the declared primary test fails. Each separate seven-of-eight path improvement has tail 0.0352, also above the retained .025 threshold. These descriptive gains neither establish the hypothesized context-specific advantage nor equivalence between futures.

At H64, matched rank 16 mean errors are 1.212 and 1.176, and exceed one in all eight histories on both paths. Short-horizon matching therefore does not certify durable preservation. Study 14 narrows the earlier interpretation: poor alternate-future transfer is not reproduced under this stable protocol. Changing batch size and learning rate together prevents assigning that difference to either change alone. The original frozen conjunction remains failed.

Increasing matched rank from 16 to 64 reduces H16 errors to 0.709/0.734 (fitted/alternate). At rank 64, however, matching worsens alternate H16 error from 0.722 to 0.734 relative to the immediate basis. H64 errors remain 1.202/1.146; eight/seven histories exceed zero-response error. Independent H16 capture remains modest: matched rank 16 captures 12.68%/12.27%, and rank 64 captures 24.49%/24.00%. This tests reuse of an H16-fitted basis at H64, without an H64-matched oracle or high-capture sufficiency test.

The response geometry distinguishes failure of direction from disappearance of magnitude (Fig. 2). For the matched rank 16 basis, mean projected/full norm rises from 0.500 to 0.912 between H16 and H64 on the fitted future, while mean cosine falls from 0.533 to 0.208. Alternate-future values change from 0.475/0.520 to 0.882/0.231 (norm/cosine). Thus a nearly full-size response need not preserve the full effect. Equation 7 holds separately within each history; inserting cohort-average norms and cosines does not reconstruct mean error. This descriptive decomposition does not identify the dynamical cause of misalignment.

## 5 DISCUSSION AND LIMITATIONS

The connected studies distinguish three assertions: memory edits cause later effects; an operator has concentrated finite-bank influence directions; and a projection preserves a relevant historical effect across future learning. The failed confirmation tests their conjunction. Horizon matching shows partial directional gains. Their context boundary varies with the continuation regime: the stable follow-up does not reproduce the earlier alternate-future failure, yet loses preservation at a longer horizon.

Table 3: CIFAR10 matched rank 16 comparisons. Errors are history means; distinct cohorts and jointly changed protocols prevent a causal regime attribution.

| | Earlier (13) | Stable (14) |
|---|---|---|
| Independent histories | 4 | 8 |
| Future batch / LR | 32/.001 | 128/.00012 |
| H16 fitted error | .594 | .838 |
| H16 alternate error | .993 | .850 |
| H64 $\Delta$acc. (pp) | $-11.34$–$-3.91$ | $+.91$–$+1.86$ |

The boundary remains conditional. All interventions use a Euclidean moment chart and need not be reachable through natural training histories. Rank and capture depend on the output metric, anchors and horizon. Moderate capture cannot supply a counterexample to sufficiency at high capture. Four histories per earlier natural-data cell and eight in the stable CIFAR10 follow-up limit inference; stable CIFAR100 remains unobserved. Approximate-basis checks cover limited estimator comparisons. Earlier cross-study comparisons confound scale and continuation protocol; deteriorating continuation limits their stable-optimization interpretation. Finally, compact archives omit completed large-model states, so native replay requires retraining.

For future optimizer-state compression studies, these results motivate reporting independent-anchor capture alongside finite-effect error, with unchanged-continuation, matched-horizon and alternate-future comparisons. Stable endpoint behavior qualifies the optimization regime; it does not certify fidelity of the much smaller intervention responses. These are evaluation recommendations grounded in the tested settings, not guarantees for other compressors. The stable follow-up qualifies baseline continuation but leaves high-capture sufficiency and cross-dataset generality unresolved. Its failed primary interaction must be retained. Completing the missing CIFAR100 cohort is future work; neither nested branches nor earlier CIFAR100 histories substitute for it.

## 6 CONCLUSION

The original low-rank causal-control conjunction fails in its frozen controlled test. Exact finite-anchor and architecture diagnostics expose loss of immediate-projection fidelity through continued learning. Horizon matching gives partial short-horizon gains. Stable CIFAR10 continuation supports gains on both futures

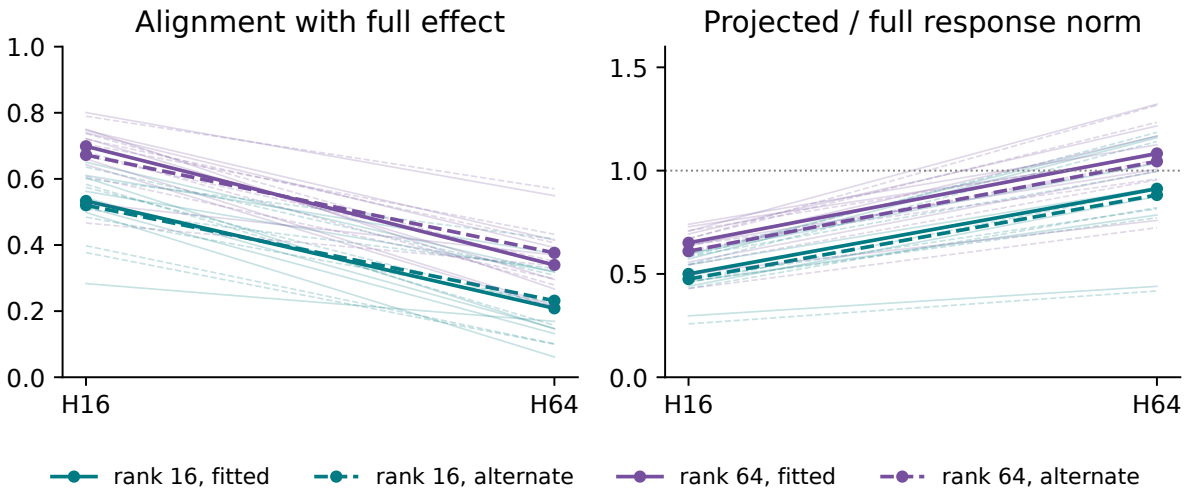



Figure 2: Stable CIFAR10 response geometry for H16-matched bases. Thick curves: eight-history means; thin curves: individual histories. Solid/dashed: fitted/alternate future. Alignment decreases from H16 to H64 while relative magnitude increases, including at rank 64.

descriptively, contradicting a general transfer-failure claim, but does not pass the declared interaction test and does not preserve H64 effects. The evidence motivates measuring independent influence capture, finite signed effects and future-context robustness separately before assigning causal sufficiency to an optimizer-memory subspace.

## AI USE STATEMENT

Generative AI assisted with hypothesis and methodology refinement, experimental implementation and audits, numerical validation design, literature discovery and summarization, result interpretation, and drafting of text, figures and reproduction artifacts. It also assisted earlier synthetic-task design. The author executed the research notebooks and supplied completed archives. Saved-array checks are distinct from independent neural-execution replication. The author is responsible for the final manuscript, claims and references.

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# CHECKLIST

1. For all models and algorithms presented, check if you include:
   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes] Sections 2–3 and Appendices A–C.
   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes] Appendix A states operator-action and storage costs.
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes] Frozen protocol code accompanies the working manuscript.

2. For any theoretical claim, check if you include:
   (a) Statements of the full set of assumptions of all theoretical results. [Not Applicable] No new theoretical result is claimed.
   (b) Complete proofs of all theoretical results. [Not Applicable] The elementary distinction is derived in Appendix A.
   (c) Clear explanations of any assumptions. [Not Applicable] Derivative assumptions are stated in Appendix A.

3. For all figures and tables that present empirical results, check if you include:
   (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [No] Plotting inputs and code are included; compact archives omit model states, so full native replay requires retraining and raw result archives.
   (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes] Section 3 and Appendix C; executable frozen settings provide additional detail.
   (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes] Section 3.4 and figure captions define independent units and plotted ranges.
   (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes] Appendix C records the matching and confirmation environments.

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
   (a) Citations of the creator If your work uses existing assets. [Yes] CIFAR, model and optimizer sources are cited.
   (b) The license information of the assets, if applicable. [No] External package licenses apply; the creator page does not supply an explicit dataset redistribution license in the material inspected. CIFAR images are not redistributed.
   (c) New assets either in the supplemental material or as a URL, if applicable. [Yes] Protocol code and plotting evidence are included.
   (d) Information about consent from data providers/curators. [Not Applicable] Existing public datasets; no new data collection.
   (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable] No new personal data collected or image content reproduced.

5. If you used crowdsourcing or conducted research with human subjects, check if you include:
   (a) The full text of instructions given to participants and screenshots. [Not Applicable] No crowdsourcing or human-subject study.
   (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable] No crowdsourcing or human-subject study.
   (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable] No crowdsourcing or human-subject study.

# A OPERATOR ALGEBRA AND ADAMW TANGENT

For fixed future batches, let $g_j = \nabla_\theta \ell_j(\theta_{j-1})$ and $G_j = \nabla^2_\theta \ell_j(\theta_{j-1})$. With elementwise operations, AdamW evolves by

$$m_j = \beta_1 m_{j-1} + (1-\beta_1) g_j, \qquad v_j = \beta_2 v_{j-1} + (1-\beta_2) g_j^2, \tag{8}$$

$$\theta_j = (1-\eta_j\lambda)\theta_{j-1} - \eta_j \frac{m_j/(1-\beta_1^{t+j})}{\sqrt{v_j/(1-\beta_2^{t+j})}+\epsilon}. \tag{9}$$

Moment-only tangents initialize $(\dot\theta_0, \dot m_0, \dot v_0) = (0, u, 0)$ and satisfy

$$\dot g_j = G_j \dot\theta_{j-1}, \tag{10}$$

$$\dot m_j = \beta_1 \dot m_{j-1} + (1-\beta_1)\dot g_j, \qquad \dot v_j = \beta_2 \dot v_{j-1} + 2(1-\beta_2) g_j \dot g_j, \tag{11}$$

$$\dot\theta_j = (1-\eta_j\lambda)\dot\theta_{j-1} + b_j \dot m_j + c_j \dot v_j, \tag{12}$$

$$b_j = -\frac{\eta_j}{(1-\beta_1^{t+j})(r_j+\epsilon)}, \qquad c_j = \frac{\eta_j m_j}{2(1-\beta_1^{t+j})(1-\beta_2^{t+j}) r_j (r_j+\epsilon)^2}, \tag{13}$$

$$r_j = \sqrt{v_j/(1-\beta_2^{t+j})}. \tag{14}$$

These expressions assume positive second-moment roots and a differentiable model/loss along the path. The first gradient tangent is zero because surgery fixes weights. The fresh horizon-matched runner differentiates all later weight/moment couplings. The clipped-gradient architecture extension uses its actual shared clipped gradient for the immediate operator; its formula should not be confused with an unqualified full-horizon derivative through clipping.

For fixed finite $H$ and a twice continuously differentiable readout, $D_H(u) = \alpha A_H u + o(\alpha^2)$ as $\alpha \to 0$. For nonzero $A_H u$, central reconstruction therefore approaches $\|A_H(I-P)u\|/\|A_H u\|$. This directional ratio has no automatic bound from capture under a different operator. For example, $A_1 = (1,0)$, $A_H = (0,1)$, $P = \mathrm{diag}(1,0)$ and $u = (1,1)/\sqrt{2}$ give full immediate capture and exact immediate preservation, but future reconstruction error one. For the single historical probe, a direction-aware rank-one basis spanning $u$ gives $Pu = u$ and preserves its intervention exactly at every horizon. It is a trivial positive control, not a fitted result or a representation of arbitrary interventions. Thus our failure concerns the tested direction-agnostic selection rules, not existence of a low-rank representation for one probe. This elementary algebra illustrates the logical gap; it does not claim that arbitrary operator pairs are realized by AdamW training or prove an empirical impossibility result.

With $\ell$ range columns and one power iteration, the fresh estimator uses $2\ell$ adjoint and $2\ell$ tangent actions plus QR/Ritz operations. A rank-$k$ capture numerator uses $k$ tangent actions; its $q$-probe denominator uses $q$ adjoint actions. Range storage is $O(p\ell + N(C-1)\ell)$, in addition to trajectory/autodifferentiation storage. This computational cost is why branch counts are workload rather than inferential sample size.

# B FROZEN GATES AND OUTCOMES

**Precursor motivation, kept outside the low-rank evidence chain.** Study 06 calibrated each direction to the same one-step class-centered-logit RMS; potency is the inverse first-moment norm required to reach that target. Across thirty confirmation tasks, the registered median of the per-task median-random/history norm ratios was 90.93 (95% task-bootstrap interval 75.95–94.57); the median best-of-eight-random/history ratio was 70.79 (58.96–82.00). In a separate within-task scalar-prediction test, relative MSE improvement over a rich comparator was 15.74% and 15.68% on two independently partitioned query views, while improvement over physical zero response was −2.18% and −2.26%. Thus strong susceptibility and beating a complicated comparator did not establish prediction better than no effect. No operator subspace $P_k$ was fit in this test, so these values neither support nor refute low-rank causal preservation. They supply motivation only and are not counted as an independent unit in Table 1.

The original confirmation requires the foundation, low-rank capture, historical alignment/privilege and causal sufficiency jointly. Discovery fails prequalification and freezes the largest candidate, rank 16. Low-rank headline

eligibility additionally requires rank at most 8. The confirmation capture gates require mean at least 0.80, task-bootstrap lower bound at least 0.70, and at least 80% of tasks with capture at least 0.60. No confirmation task clears the last threshold.

Table 4: Selected frozen confirmation gates and outcomes ($n = 30$ tasks). Intervals are the archived 95% task-bootstrap intervals. Failed conjunctions remain failed.

| Measure | Mean | Interval | Declared comparison |
|---|---|---|---|
| Independent capture | .3841 | [.3669,.4023] | Mean$\geq$ .80, lower$\geq$ .70 |
| Historical squared projection fraction | .0443 | [.0381,.0504] | Mean$\geq$ .15 |
| Alignment excess over matched random | .0443 | [.0380,.0504] | Lower$\geq$ .05 |
| H16 projected/full norm | .6891 | [.6179,.7641] | Mean$\geq$ .70, lower$\geq$ .60 |
| H16 projected/full cosine | .5542 | [.4647,.6356] | Lower$\geq$ .75 |
| H16 remainder/full norm | .8416 | [.7677,.9159] | Mean$\leq$ .40, upper$\leq$ .55 |
| H16 additivity error | .00118 | [.00104,.00133] | Mean$\leq$ .35 |

The same causal gates apply separately to both signed baseline-relative effects, in addition to the central effects. Both signs fail. Foundation qualification passes; median final and recipient accuracy are 78.78% and 77.15%. Controlled-task inference does not establish shared bases across tasks: rank transfers, while each task fits its own basis.

# C TRAINING AND MEASUREMENT DETAILS

Table 5: Recorded hyperparameters. All use AdamW betas (.9,.999), epsilon $10^{-8}$, fixed 0.002-relative first-moment displacement, no dropout or batch normalization.

| Study | Training updates | Train batch | Learning rate | Weight decay | Future batch |
|---|---|---|---|---|---|
| 07 controlled | 500 | 128 | .0012 | .0001 | 128 |
| 10 exact CIFAR | 2,000 | 128 | .0012 | .0001 | 128 |
| 11 extension C10/C100 | 8,000/12,000 | 128 | .0012 peak, scheduled | .01 | 128 |
| 12 small matching | Reuses 10 | 128 | .0012 | .0001 | 128 |
| 13 fresh matching | 8,000 | 128 | .001 constant | .0005 | 32 |
| 14 stable follow-up | 8,000 | 128 | .001 train/.00012 future | .0005 | 128 |

Study 11 uses 500 warmup steps followed by cosine decay to 0.1 times peak learning rate and gradient clipping at norm 1. Its first common calibration batch contains 256 unaugmented images. Fresh study 13 uses no schedule or clipping; its first common calibration batch contains 128 images. Study 12 uses the first 128 images of its reused calibration role. Later natural-data batches use deterministic crop/flip augmentation and replacement sampling. Fresh-study pixel scaling is $(x/255 - .5)/.25$ on both datasets.

Study 13 fitting/held-out/first-update roles are disjoint within the reserved 4,096-image calibration partition. Fit64 is a prefix of fit 256. For CIFAR100, fit 64 contains at most 64 classes, chosen by randomized balanced sampling. Fitting and capture use logits, not class labels; the first-update loss uses calibration labels. Response images and endpoint images are disjoint official test subsets. Endpoints never select seeds or tune the protocol. All histories are retained.

Recorded execution uses Google Colab with NVIDIA A100-SXM4-40GB, Python 3.13.15 and NumPy 2.1.3. The controlled confirmation uses PyTorch 2.11.0+cu128, CUDA 12.8 and cuDNN 91900; the matching archives record PyTorch 2.11.0+cu130, CUDA 13.0 and cuDNN 92700. Study 12 records torchvision 0.26.0+cu130. Deterministic algorithms are enabled and TF32 is disabled. The current saved-array review is a separate CPU calculation and does not rerun those GPU experiments.

# D COMPLETED STABLE CIFAR10 COHORT

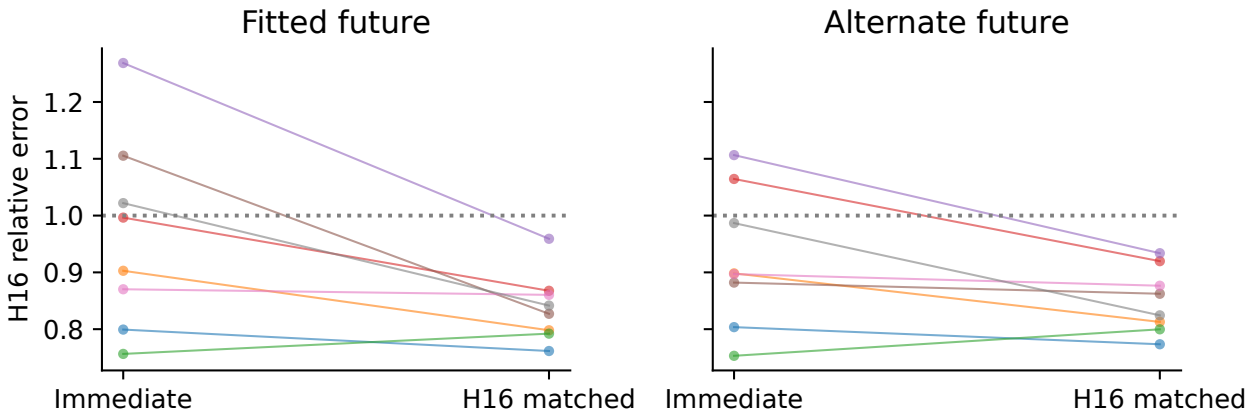



Figure 3: Stable CIFAR10 follow-up: H16/rank 16 errors, eight independent histories. Lines pair immediate and matched bases. Matching helps both futures descriptively; the declared interaction test fails. Dotted line: zero-response prediction.

Study 14 reports all eight completed CIFAR10 histories, 14000–14007. The planned stable CIFAR100 replication was unfinished at the reporting cutoff and is not analyzed; earlier CIFAR100 studies use distinct protocols. The source archive's 1,000 manifest payloads pass byte-size, SHA256 and ZIP CRC checks. Included anonymized branch logits support CPU recomputation of response errors; baseline scores are archived endpoint aggregates, not an independent full-test replay. The primary interaction is immediate-minus-matched error on path 0 minus the same gain on path 1. Positive values favor the fitted future. The original two-cohort per-dataset threshold .025 is preserved.

Table 6: Stable follow-up H16/rank 16 matching gain by independent history. Gains are relative-error differences, not accuracy gains.

| History | Fitted gain | Alternate gain | Interaction |
|---|---|---|---|
| 14000 | 0.0377 | 0.0301 | 0.0076 |
| 14001 | 0.1047 | 0.0855 | 0.0192 |
| 14002 | -0.0357 | -0.0466 | 0.0109 |
| 14003 | 0.1288 | 0.1448 | -0.0159 |
| 14004 | 0.3095 | 0.1727 | 0.1367 |
| 14005 | 0.2784 | 0.0197 | 0.2587 |
| 14006 | 0.0099 | 0.0205 | -0.0106 |
| 14007 | 0.1804 | 0.1624 | 0.0180 |

Table 7: Stable follow-up: matched-basis mean relative errors and H64 histories above zero-response error. Rank 64 is a secondary diagnostic.

| Future / rank | H16 error | H64 error | H64 error> 1 | H16 capture |
|---|---|---|---|---|
| Fitted /16 | .838 | 1.212 | 8/8 | 12.68% |
| Alternate /16 | .850 | 1.176 | 8/8 | 12.27% |
| Fitted /64 | .709 | 1.202 | 8/8 | 24.49% |
| Alternate /64 | .734 | 1.146 | 7/8 | 24.00% |

The extended saved-array audit checks both signs, ranks 16/64 and horizons 16/64. Maximum signed-error discrepancy from central error is .00325; the largest full-effect even/odd ratio is 1.85%, and additivity error is 1.66%. Minimum full-effect RMS is $6.74 \times 10^{-5}$, above the frozen $2 \times 10^{-5}$ floor. Maximum archived derivative prediction error at H16/full and half amplitude is 1.31%. Recomputing full versus twice-half-amplitude historical effects gives maximum relative discrepancies of 1.37% at H16 and 2.42% at H64. This supports approximate amplitude linearity of the full probe, but does not validate unrun half-amplitude projected branches. These are finite numerical checks, not independent neural-execution replication. Maximum baseline-relative endpoint accuracy and NLL changes over all 400 branches are 0.01013 percentage points and $1.75 \times 10^{-5}$. Thus observable logit responses do not establish practical endpoint benefit.

All histories satisfy the frozen stability and numerical eligibility checks. The primary sign test has 6/8 positive interactions ($p = .14453125$); failed tests are not converted into claims of equality or absence. The wider-sketch checks only assess two histories and finite fitting banks. The follow-up changes continuation batch size and learning rate jointly, so it cannot causally identify the source of study 13's deterioration. Earlier completed CIFAR100 results remain labeled study 11/13 evidence and do not represent stable-continuation replication.

## E ADDITIONAL DIAGNOSTICS

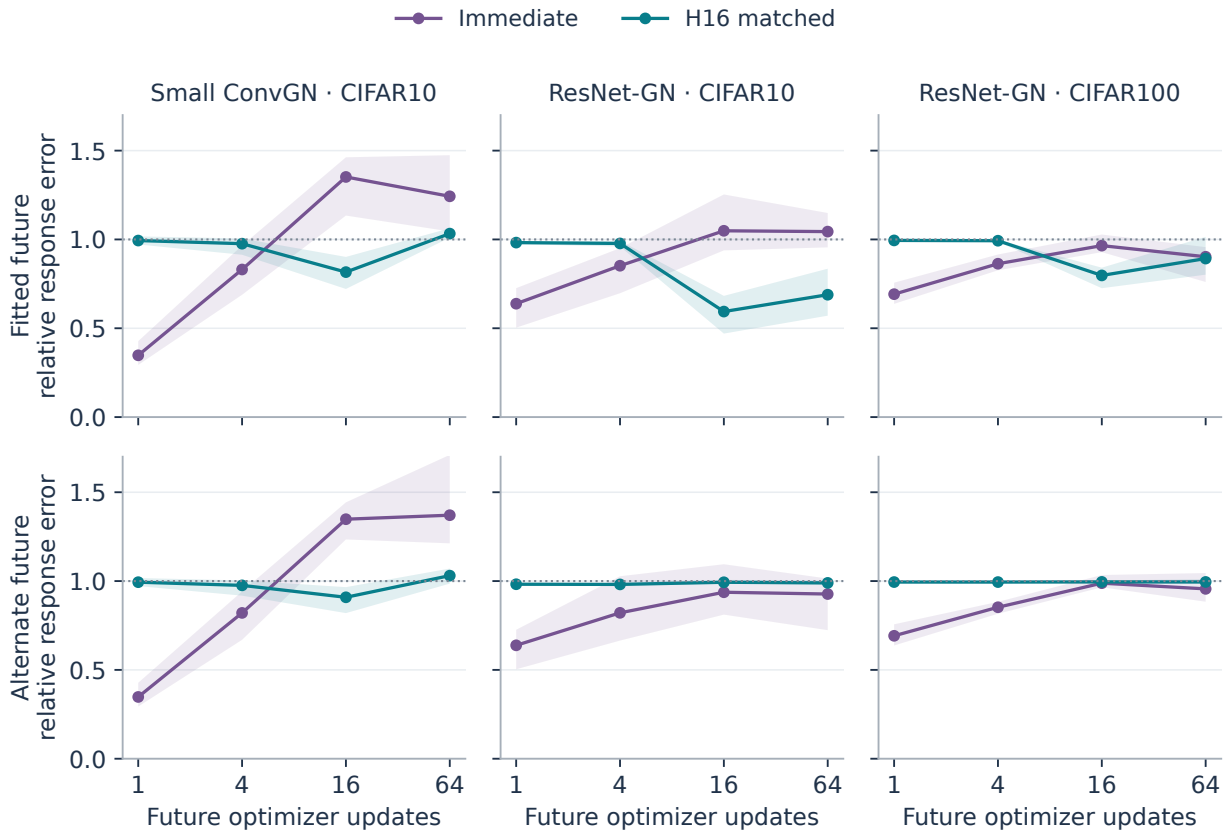



Figure 4: Study 12/13: rank 16 reconstruction across horizons. Lines are four-history means; shaded areas are minimum–maximum ranges, not confidence intervals. The matched operator uses sixteen updates. The small and fresh cohorts differ in model, training length, fitting bank and future batch size; their contrast is not a model-size ablation.

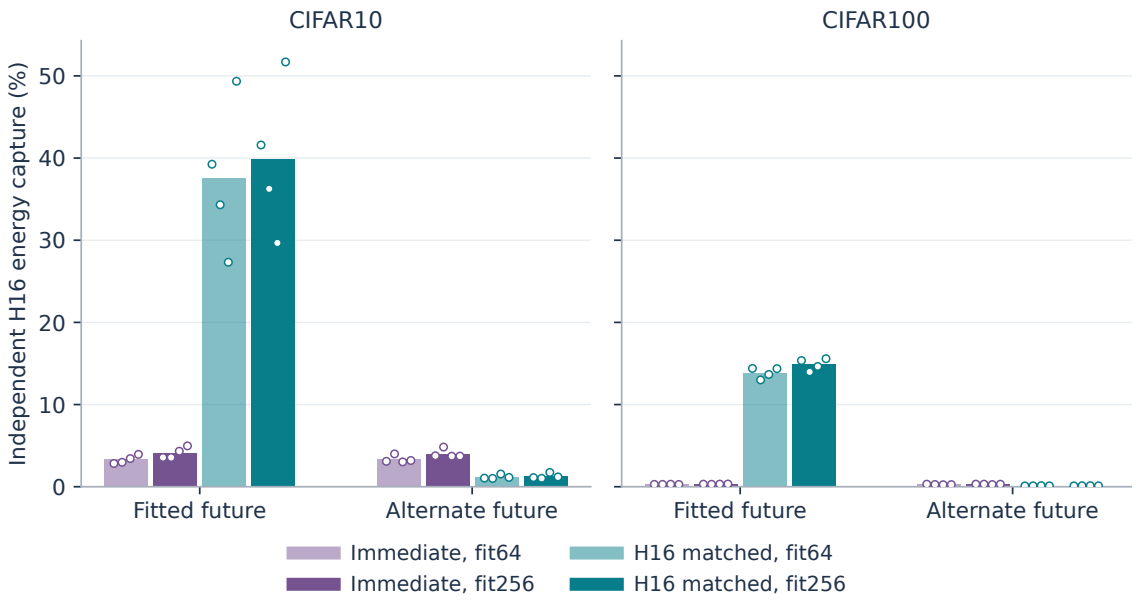



Figure 5: Independent operator capture in the fresh study. Bars are four-history means and points are seed estimates. Denominators use 128 independent Rademacher trace probes. Low rank 16 capture limits conditional sufficiency claims.

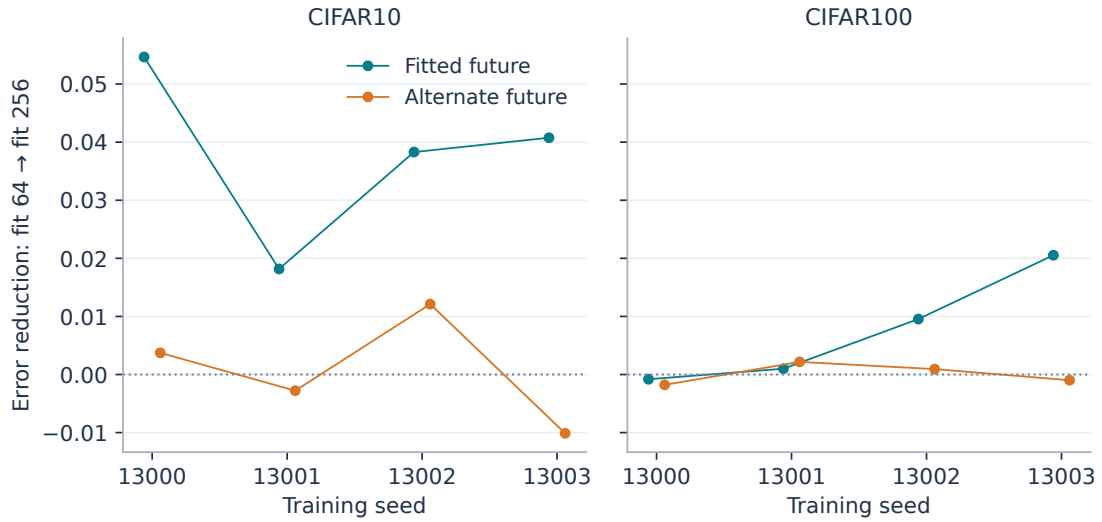



Figure 6: Within-history error reduction from fit 64 to fit 256. Positive values mean the larger bank helps. Image-support expansion yields little alternate-future improvement.

# F BASELINE CONTINUATION AND REPRODUCIBILITY BOUNDARY

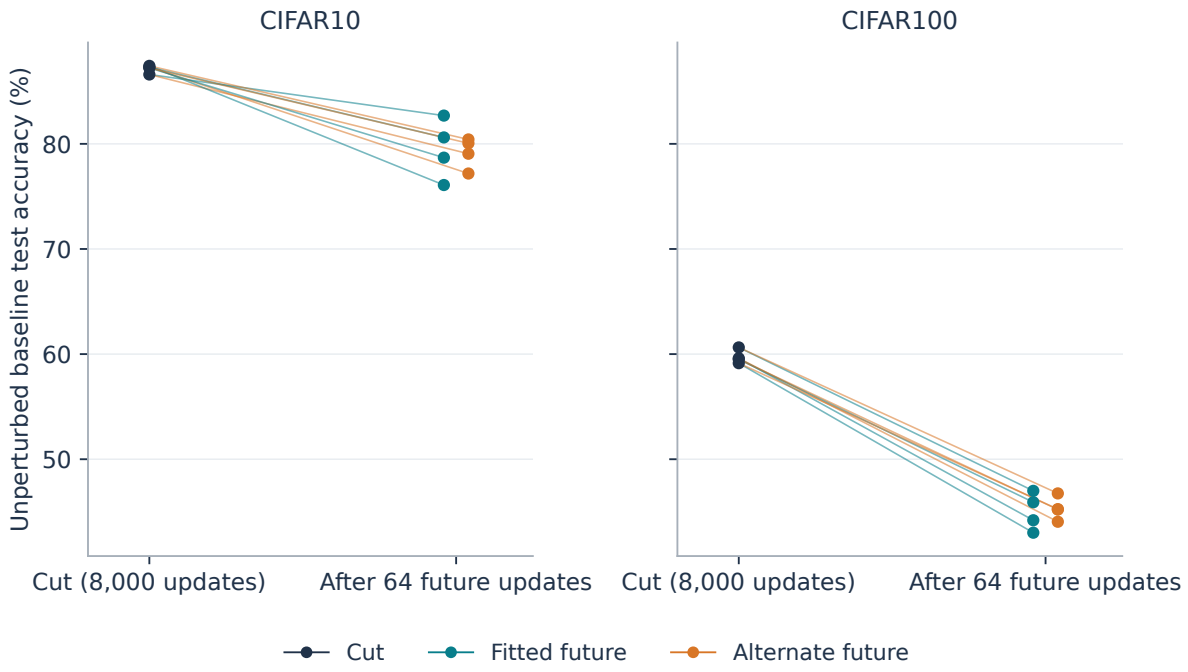



Figure 7: Unperturbed study 13 continuation loses substantial accuracy, separately from the much smaller intervention endpoint changes. Continuation switches from training batch 128 to batch 32 after the shared calibration step. The batch-size change is a possible contributor, not an isolated causal explanation.

The audit verifies 496 Stage12 and 1,588 Stage13 manifest payloads, four/eight completed units, and 200/656 branches. Recomputed central and separate signed response metrics, numerical residuals and fresh capture arithmetic agree with saved results. Study 14 adds eight completed CIFAR10 histories and 400 branches; all 256 central projection-error comparisons recompute exactly from saved logits. Baseline accuracy/NLL drift uses archived aggregate scores rather than a retained full-test prediction replay. Earlier-stage summaries retain their own saved-data audit provenance. The accompanying source/evidence bundle contains plotting inputs, computations and frozen protocol code; the compact large-model archives omit completed weights and moments. Reproduction of native surgery therefore requires retraining, while figures can be regenerated from the included tables without a GPU.

CIFAR data are obtained from their creator's public distribution and are not redistributed with the bundle. Models are project implementations of cited architectures. PyTorch/torchvision are external dependencies; their licenses remain with their distributions. No newly collected human-subject data are used. Full dependency and provenance records accompany the code.